

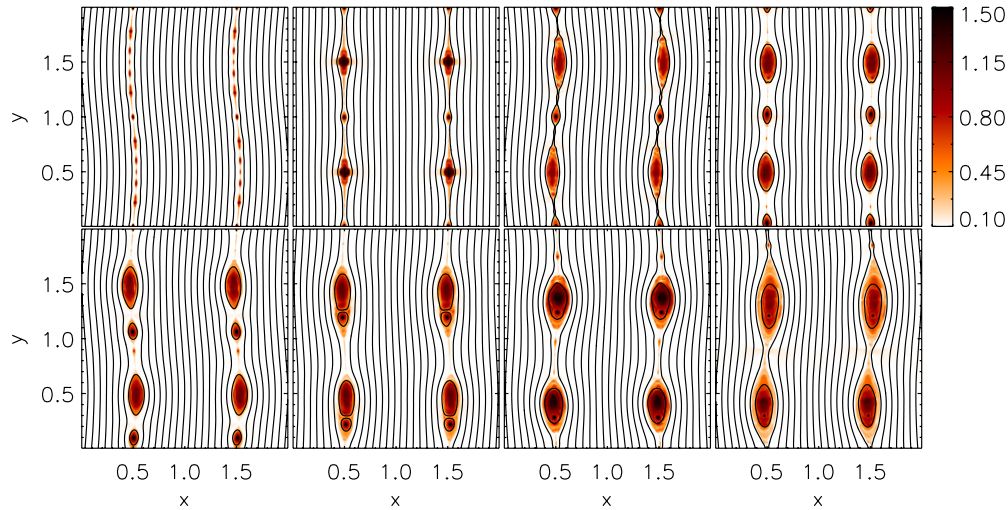


Figure 15. Time evolution of the pressure profiles along with sample magnetic field lines, for the current sheet problem. Temporal snapshots refer to $t = 0.5, 1, 1.5, 2$ (upper four) and $t = 2.5, 3, 3.5, 4$ (lower four).

(A color version of this figure is available in the online journal.)

number of processors N_{CPU} for the three- and five-level AMR computations as well as for fixed uniform grid runs carried out at the equivalent resolution of 2048×1024 zones. Here, T_1 is the same reference constant for all calculations and is equal to the (inferred) running time of the single processor static mesh computation while $T_{N_{\text{CPU}}}$ is the execution time measured with N_{CPU} processors. The scaling reveals an efficiency (defined as S/N_{CPU}) larger than 0.8 for less than 256 processors with the three- and five-level computations being, respectively, eight to nine and four to five times faster than the fixed grid approach. The number of blocks on the finest level is maximum at the end of integration and is slightly larger for the three-level run (1058 versus 835). This result indicates that using fewer levels of refinement with larger grid ratios can be more efficient than introducing more levels with consecutive jumps of 2, most likely because of the reduced integration cost due to the missing intermediate levels and the decreased overhead associated with coarse-fine level communication and grid generation process. Efficiency quickly drops when the number of CPU tends to become, within a factor of between two and three, comparable to the number of blocks.

5.4. Current Sheet

The current sheet problem proposed by Gardiner & Stone (2005) and later considered by Fromang et al. (2006) in the AMR context is particularly sensitive to numerical diffusion. The test problem follows the evolution of two current sheets, initialized through a discontinuous magnetic field configuration. Driven solely by numerical resistivity, reconnection processes take place, making the resulting solution highly susceptible to grid resolution.

The initial condition is discretized onto a Cartesian 2D grid $x, y \in [0, 2]$, with 64×64 zones at the coarse level. The fluid has uniform density $\rho = 1$ and thermal pressure $P = 0.1$. Its bulk flow velocity $\mathbf{v}$ is set to zero, allowing only for a small perturbation in $v_x = v_0 \sin(\pi y)$, where $v_0 = 0.1$. The initial magnetic field has only one non-vanishing component in the vertical direction,

$$B_y = \begin{cases} -B_0 & \text{if } |x - 1| \leq 0.5, \\ B_0 & \text{otherwise,} \end{cases} \quad (61)$$

where $B_0 = 1$, resulting in a magnetically dominated configuration. Boundaries are periodic and the integration terminates at $t = 4$. We activate refinement whenever the maximum between the two error norms (given by Equation (50)) computed with the specific internal energy and the y -component of the magnetic field exceeds the threshold value $\chi_r = 0.2$. In order to filter out noise between magnetic island we set $\epsilon = 0.05$. We allow for four levels of refinement and carry out integration with the Roe Riemann solver, the improved WENO reconstruction, and a CFL number of 0.6.

In Figure 15, we show temporal snapshots of pressure profiles for $t = 0.5, 1.0, 1.5, 2, 3.5$, and 4.0 , along with sample magnetic field lines. Since no resistivity has been specified, the elongated current sheets are prone to tearing instability as numerical resistivity is small with respect to Biskamp's criterion (Biskamp 1986). Secondary islands, “plasmoids,” promptly form and propagate parallel to the field in the y -direction. As reconnection occurs, the field is dissipated and its magnetic energy is transformed into thermal energy, driving Alfvén and compressional waves which further seed reconnection (Gardiner & Stone 2005; Lee & Deane 2009). Due to the dependency of numerical resistivity on the field topology, reconnection events are most probable at the nodal points of v_x . At the later stages of evolution, the plasmoids eventually merge in proximity of the anti-nodes of the transverse speed, forming four larger islands, in agreement with the results presented in Gardiner & Stone (2005). A close-up on the bottom left island is shown in Figure 16, at the end of integration. The refinement criterion on the second derivative of thermal pressure, as suggested by Fromang et al. (2006), efficiently captures the island features of the solution.

5.5. Three-dimensional Rayleigh–Taylor Instability

In this example, we consider the dynamical evolution of two fluids of different densities initially in hydrostatic equilibrium. If the lighter fluid is supporting the heavier fluid against gravity, the configuration is known to be subject to the Rayleigh–Taylor instability. Our computational domain is the box spanned by $x, z \in [-1/2, 1/2]$, $y \in [-3/2, 1/2]$ with gravity pointing in the negative y -direction, i.e., $\mathbf{g} = (0, -1, 0)$. The two fluids are separated by an interface initially lying in the xz -plane at $y = 0$,

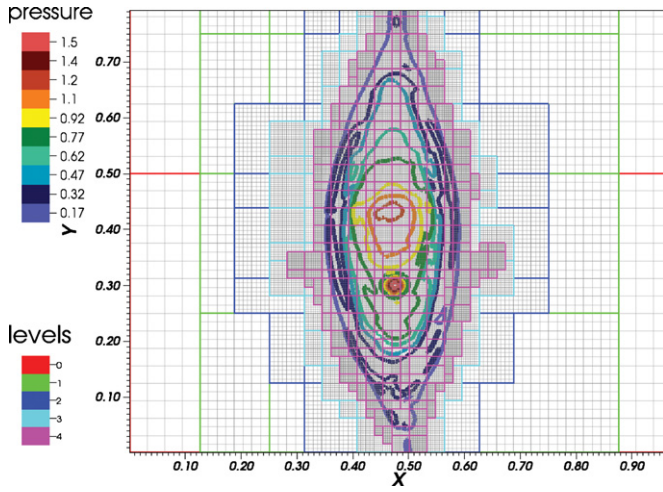


Figure 16. Close-up of the bottom left island at $t = 4$ for the current sheet problem. Pressure contours along with the refinement levels and grid are shown. (A color version of this figure is available in the online journal.)

with the heavy fluid $\rho = \rho_h$ at $y > 0$ and $\rho = \rho_l$ at $y < 0$. In the current example we employ $\rho_h = 4$, $\rho_l = 1$ and specify the pressure from the hydrostatic equilibrium condition:

$$p(y) = \frac{100}{\gamma} - \rho y, \quad (62)$$

so that the sound crossing time in the light fluid is 0.1. The seed for instability is set by perturbing the velocity field at the center of the interface:

$$v_y = -\frac{\exp[-(5r)^2]}{10 \cosh(10y)^2}, \quad (63)$$

where $r = \sqrt{x^2 + z^2}$. We assume a constant magnetic field $\mathbf{B} = (B_x, 0, 0)$ parallel to the interface and oriented in the

x -direction with different field strengths, $B_x = 0$ (hydro limit), $B_x = 0.2B_c$ (moderate field), and $B_x = 0.6B_c$ (strong field). Here, $B_c = \sqrt{(\rho_h - \rho_l)gL}$ is the critical field value above which instabilities are suppressed (Stone & Gardiner 2007). We use the PPM method with the Roe Riemann solver and a Courant number $C_a = 0.45$. The base grid has $16 \times 32 \times 16$ cells and we perform two sets of computations using (1) four refinement levels with a grid spacing ratio of 2 and (2) two refinement levels with a grid jump of 4, in both cases achieving the same effective resolution ($256 \times 512 \times 256$). Refinement is triggered using the second derivative error norm of density and a threshold value $\chi_{\text{ref}} = 0.5$. Periodic boundary conditions are imposed at the x - and z -boundaries while fixed boundaries are set at $y = -3/2$ and $y = 1/2$.

Results for the unmagnetized, weakly, and strongly magnetized cases are shown at $t = 3$ in Figure 17. In all cases, we observe the development of a central mushroom-shaped finger. Secondary instabilities due to Kelvin–Helmholtz modes develop on the side and small-scale structures are gradually suppressed in the direction of the field as the strength is increased. Indeed, as pointed out by Stone & Gardiner (2007), the presence of a uniform magnetic field has the effects of reducing the growth rate of modes parallel to it, although the interface still remains Rayleigh–Taylor unstable in the perpendicular direction due to interchange modes. As a net effect, the evolution becomes increasingly anisotropic as the field strengthens.

Parallel performance is plotted, for the moderate field case, in Figure 18 where we measure the speed-up factors of the four- and two-level computations versus the number of CPUs. Here, the speed-up is defined by $S = T_1/T_{N_{\text{CPU}}}$, where T_1 is the inferred running time relative to the four-level computation on a single processor. The two-level case shows improved performance over the four-level calculation both in terms of CPU cost as well as parallel efficiency S/N_{CPU} . Indeed the relative gain between the two cases approaches a factor of two for an increasing number of CPUs. Likewise, the efficiency of the fewer level case remains above $\gtrsim 0.8$ up to 512 processors and drops to ~ 0.61 (versus

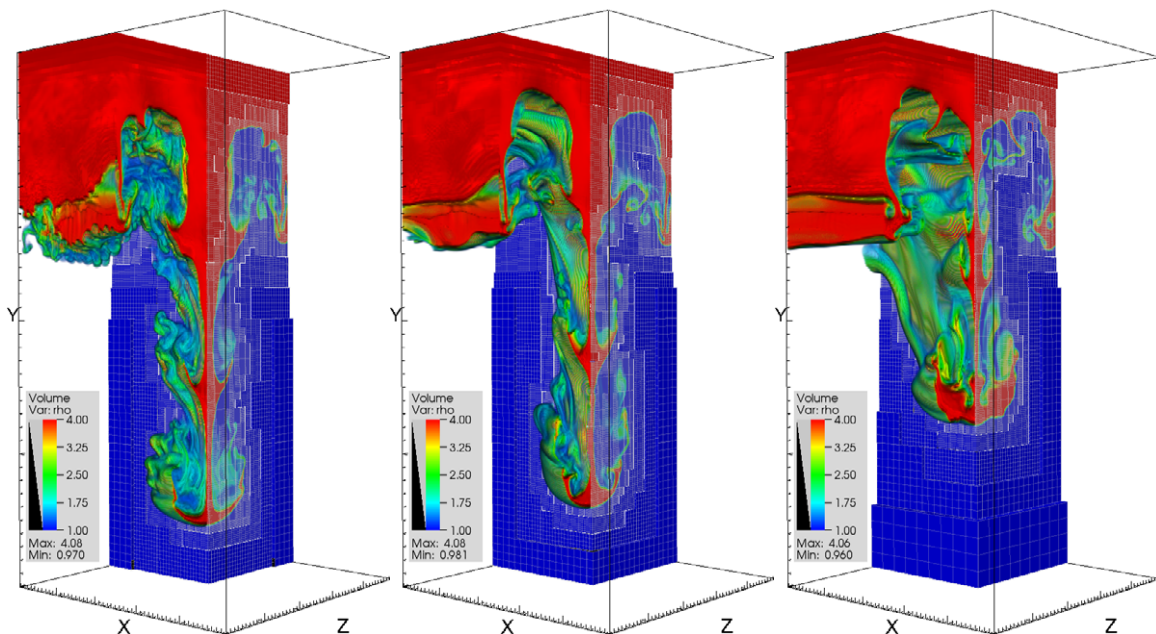


Figure 17. Three-dimensional view of the Rayleigh–Taylor instability problem showing density at $t = 3$ for the unmagnetized (left panel), weakly magnetized (middle panel), and strongly magnetized case (right panel) using four levels of refinement. In each panel, we also show a sliced box emphasizing the grid refinement ratios. (A color version of this figure is available in the online journal.)

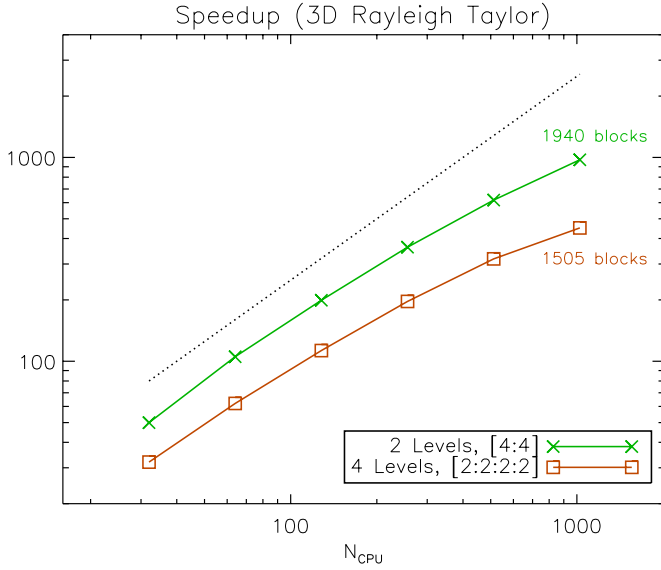


Figure 18. Parallel speed-up vs. number of processors N_{CPU} for the 3D Rayleigh–Taylor problem computed with two (green crosses) and four (red squares) refinement levels between $t = 2$ and $t = 2.25$. The dotted line gives the ideal scaling. The number of blocks on the finest level at $t = 2$ is printed above and below the corresponding curves.

(A color version of this figure is available in the online journal.)

~ 0.44 for the four-level case) at the largest number of employed cores (1024), which is less than the number of blocks on the finest grid.

5.6. Two-dimensional Shock–Cloud Interaction

The shock–cloud interaction problem has been extensively studied and used as a standard benchmark for the validation of MHD schemes and intercode comparisons (see Dai & Woodward 1994; Balsara 2001a; Lee & Deane 2009, and references therein). In an astrophysical context, it addresses the fundamental issue of the complex morphology of supernova remnants as well as their interaction with the interstellar medium. Energy, mass, and momentum exchange leading to cloud crushing strongly depends on the orientation of the magnetic field and the resulting anisotropy of thermal conduction.

Following Orlando et al. (2008), we consider the 2D Cartesian domain $x \in [-4, 4]$, $y \in [-1.4 \times 6.6]$ (in parsecs) with the shock front propagating in the positive y -direction and initially located at $y = -1$. Ahead of the shock, for $y > -1$, the hydrogen number density n_1 has a radial distribution of the form:

$$n_1 = n_a - \frac{n_a - n_c}{\cosh[\sigma(r/r_{\text{cl}})^\sigma]}, \quad r = \sqrt{x^2 + y^2}, \quad (64)$$

where $n_c = 1 \text{ cm}^{-3}$ is the hydrogen number density at the cloud’s center, $n_a = 0.1n_c$ is the ambient number density, $r_{\text{cl}} = 1 \text{ pc}$ is the cloud’s radius ($\sigma = 10$), and r is the radial distance from the center of the cloud. The ambient medium has a uniform temperature $T_a = 10^4 \text{ K}$ in pressure equilibrium with the cloud, with a thermal pressure of $p_1 = 2k_B n_a T_a$ where k_B is the Boltzmann constant (we assume a fully ionized gas). Downstream of the shock, density, and transverse components of the magnetic field are compressed by a factor of $n_2/n_1 =$

$(\Gamma + 1)/(\Gamma - 1)$ while pressure and normal velocity are given by

$$p_2 = 2n_2 k_B T_s, \quad v_{y,2} = \sqrt{\frac{2}{\Gamma - 1} \frac{2k_B T_s X}{m_H}}, \quad (65)$$

where $X = 1/1.26$ is the hydrogen mass fraction and $T_s = 4.7 \times 10^6 \text{ K}$ is the post-shock temperature. The normal component of the magnetic field (B_y) remains continuous through the shock front.

We perform two sets of simulations with a magnetic field strength of $|\mathbf{B}| = 1.31 \mu\text{G}$, initially parallel (case Bx4) or perpendicular (case By4) to the shock front. Adopting the same notations as Orlando et al. (2008), we solve in each case the MHD equations with (TN) and without (NN) thermal conduction effects for a total of four cases. Thermal conductivity coefficients along and across the magnetic field lines (see Equation (6)) are given, in cgs units, by

$$k_{\parallel} = 9.2 \times 10^{-7} T^{5/2}, \quad k_{\perp} = 5.4 \times 10^{-16} \frac{n_{\text{H}}^2}{\sqrt{T} B^2}. \quad (66)$$

The resolution of the base grid is set to 32^2 points and five levels of refinement are employed, yielding an equivalent resolution of 1024×1024 . Upwind characteristic tracing (Section 3.2) together with PPM reconstruction (Equation (33)) and the HLLD Riemann solver are chosen to advance the equations in time. Open boundary conditions are applied on all sides, except at the lower y -boundary where we keep constant inflow values. The CFL number is $C_a = 0.8$. Refinement is triggered upon density with a threshold of $\chi_{\text{ref}} = 0.15$.

In Figures 19 and 20, we show the cloud evolution for the four different cases at three different times $t = 1, 2, 3$ (in units of the cloud crushing time $\tau_{\text{cc}} = 5.4 \times 10^3 \text{ yr}$). Our results are in excellent agreement with those of Orlando et al. (2008), confirming the efficiency of thermal conduction in suppressing the development of hydrodynamic instabilities at the cloud borders. The topology of the magnetic field can be quite effective in reducing the efficiency of thermal conduction. For a field parallel to the shock front (TN-Bx4), the cloud’s expansion and evaporation are strongly limited by the confining effect of the enveloping magnetic field. In the case of a perpendicular field (TN-By4), on the contrary, thermal exchange between the cloud and its surroundings becomes more efficient in the upwind direction of the cloud and promotes the gradual heating of the core and the consequent evaporation in a few dynamical timescales. In this case, heat conduction is strongly suppressed laterally by the presence of a predominantly vertical magnetic field.

5.7. Radiative Shocks in Stellar Jets

Radiative jet models with a variable ejection velocity are commonly adopted to reproduce the knotty structure observed in collimated outflows from young stellar objects. The time variability leads to the formation of a chain of perturbations that eventually steepen into pairs of forward/reverse shocks (the “internal working surfaces”; see, for instance, de Colle & Raga 2006; Raga et al. 2007; Teşileanu et al. 2009) which are held responsible for the typical spectral signature of these objects. While these internal working surfaces travel down the jet, the emission from post-shocked regions occurs on a much smaller spatial and temporal scale, thus posing a considerable challenge even for AMR codes.